

T_0 Space (Kolmogorov)

For distinct $x, y \in X$:

$\exists$ open set U containing

one but not the other

(Can distinguish points)

T_1 Space

For distinct $x, y \in X$:

$\exists$ open set U_x with

$x \in U_x, y \notin U_x$ and

$\exists$ open set U_y with

$y \in U_y, x \notin U_y$

T_2 Space (Hausdorff)

For distinct $x, y \in X$:

$\exists$ disjoint open sets

$U \ni x$ and $V \ni y$ with

$U \cap V = \emptyset$